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The Road to Healthier Diet: Benefits of the "Five Vertical and Seven Horizontal" National Trunk Highway System in China☆

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ABSTRACT

Billions of people worldwide still lack access to healthy diets, with a high concentration in rural areas of developing countries. This paper examines how major transportation investments can improve dietary quality among rural households, leveraging the staggered rollout of the "Five Vertical and Seven Horizontal" National Trunk Highway System (5V7H), the country's largest expressway network completed by 2007. Using a staggered difference-in-differences design, we find that the 5V7H access increases the Dietary Diversity Score (DDS) and Chinese Healthy Eating Index (CHEI) of rural residents by 0.326 and 2.197 points, respectively. These benefits are more prominent among households with more children, access to refrigerators, or meal preparers with better nutrition knowledge, and less so among households with more diversified agricultural production. We further show that the 5V7H connection improves rural residents' dietary quality primarily through demand-side channels, including promoting off-farm employment, raising household income, and enhancing dietary literacy. In contrast, the contributions of supply-side channels, such as improved market access or lower food prices, are modest. Overall, our findings highlight the benefits of large-scale transportation infrastructure in facilitating the transition to healthier diets in rural areas of developing countries.

1. Introduction

It is estimated that 3.1 billion people worldwide still lack access to healthy diets, with a high concentration in rural areas of developing countries FAO et al., 2023. Low levels of income (Bai et al., 2021; Hirvonen et al., 2020; Tian and Yu, 2015), taste preferences for unhealthy foods (Dolislager et al., 2022; Mekonnen et al., 2021; Zhao and Yu, 2019), food cultures (Atkin, 2016), food price (McCullough et al., 2022) and high time cost of accessing healthy food (Dolislager et al., 2022) have been blamed for such a high

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concentration. This paper proposes another possible reason, which has been much less explored in the literature, inadequate transportation infrastructure in rural areas of developing countries (Asher and Novosad, 2020).

We seek to examine the impact of transportation on the dietary quality of rural residents in developing countries using China as a valuable setting for a couple of reasons. First, China has demonstrated remarkable progress towards the "Zero Hunger" Sustainable Development Goal. According to the National Health Commission (NHC, 2021), the prevalence of malnutrition in rural China declined to 4.1% in 2020. Second, the staggered rollout of China's first large-scale transportation infrastructure project since its founding in 1949, the "Five Vertical and Seven Horizontal" National Highway System (or the 5V7H hereafter),[1] provides a quasi-experiment to explore the causal relationship between transportation accessibility and dietary quality.

We leverage the staggered rollout of the 5V7H and take a staggered difference-in-differences (DID) approach. We find that the 5V7H connection had a statistically significant positive impact on the dietary quality of rural residents in China. Compared to their rural peers without connection to the 5V7H (counties that did not contain any 5V7H entrance or exit), rural residents connected to the 5V7H (counties that contained at least one 5V7H entrance or exit) consumed 0.326 more food groups in the Dietary Diversity Score (DDS) and scored 2.197 points higher on the Chinese Healthy Eating Index (CHEI). Given that the average DDS is 4.62 and the average CHEI is 50.41 at baseline, these estimates imply increases of approximately 7.1 percent and 4.4 percent, respectively. In other words, these estimated effects imply that compared with their rural peers without the 5V7H connection, rural residents with the connection not only exhibit more dietary diversity (an increase of 0.33 in the number of food groups consumed over the three-day reference period), but also have diets that are more aligned with the recommended dietary pattern. The improvement in CHEI was manifested in the increased intake of recommended food groups (such as fruits, aquatic products, nuts, and milk) whereas reduced oil intake, which contributes to a reduction in obesity prevalence among rural residents. Our results are robust to parallel pre-trend test, sensitivity test of alternative distance cutoffs, controlling for confounding factors as well as alternative estimation specifications. Further analyses reveal the 5V7H connection improves rural residents' dietary quality primarily through demand-side factors (such as off-farm employment, increased household income, and improved dietary literacy) rather than supply-side factors. Moreover, we find that low-income households benefit mainly through increased income whereas high-income households benefit through improved dietary literacy. We also find the dietary quality benefits are more prominent among households with more children, refrigerators, meal preparers with better nutrition knowledge, but less so for households with more diversified agricultural production.

The contribution of this study is threefold. First, we contribute to the literature on how transportation affects the well-being of rural residents. Previous studies have examined the impact of transportation infrastructure, especially rural roads, on farmers' income and welfare, as well as agricultural productivity (Aggarwal, 2021, 2018; Asher and Novosad, 2020; Lu et al., 2022; Rammelt and Leung, 2017; Zhang et al., 2023). This paper focuses on highway, namely, the "Five Vertical and Seven Horizontal" National Highway System in China. Moreover, we explored the potential interaction between the 5V7H connection and rural roads in affecting the dietary quality of rural residents.

Second, this paper contributes to the discussion on role of demand- or supply-side factors in improving dietary quality. Some studies blame such supply side factors as the prevalence of "food deserts" for low dietary quality (Bitler and Haider, 2010; Lewis et al., 2005; Powell et al., 2007). However, Allcott et al. (2019) reveal that eliminating income disparities (an important demand side factor) reduces nutritional inequality by only about 10 percent, the remaining 90 percent is attributable to differences in other demand side factors. This paper finds that the 5V7H connection improves rural residents' dietary quality primarily through demand-side factors (such as off-farm employment, increased household income, and improved dietary literacy) rather than supply-side factors. Moreover, we find low-income households benefit mainly through increased income whereas high-income households via improved dietary literacy.

Finally, this paper is also related to the literature on agricultural production diversity and dietary quality. While production diversification may bolster nutritional outcomes for smallholder households, it could simultaneously diminish their consumption of externally sourced foods (Jones et al., 2014; Powell et al., 2015; Sibhatu et al., 2015). We supplement this strand of literature by examining the extent to which rural residents with different production diversity exhibit different improvement in dietary quality after the 5V7H connection.

The remainder of this paper is organized as follows. Section 2 briefly describes the 5V7H project in China. Section 3 outlines the conceptual framework. Section 4 describes the data. Section V details the empirical strategy. Section 6 presents empirical results. Section 7 examines heterogeneity and the last section concludes.

2. The 5V7H project in China

In response to the country's increasing demand for transportation infrastructure, China's Ministry of Transportation proposed the five north-south (vertical) highways and seven east-west (horizontal) highways (5V7H) project in the late 1980s and officially released the construction plan in 1993. It is stated in the plan that 12 trunk highways with a total mileage of over 35,000 km will be built by 2020 to connect all important cities, industrial centers, transportation hubs, and major land crossings in China, and to cover all megacities with a population of more than one million and most medium-sized cities with a population over 500,000. As it turned out,

[1] The 5V7H project was proposed by China's Ministry of Transport in the late 1980s to build five longitudinal highways from north to south and seven horizontal highways from east to west within 30 years to connect important cities and industrial centers with populations of 500,000 or more along the lines, as well as land ports. The construction of 5V7H began in 1993. By 2004, 91% of its mainlines were completed. By 2007, the entire 5V7H system was in operation, with a total mileage of approximately 35,000 km (Faber, 2014).

the 5V7H construction plan was accomplished ahead of schedule in 2007, with most of its construction works undertaken between 1998 and 2004. According to Li and Shum (2001), by the end of 1997, 4771 km (or nearly 14%) of the total planned length of highways had been completed. By 2004, 91% of the major lines had been completed, making it the largest of its kind since the founding of the People's Republic of China.

The 5V7H project was built in a staggered manner given the way it was planned, financed and implemented. The project was planned by the central government but jointly financed by local (provincial and county) governments (roughly 70 percent), the central government (10 to 15 percent) and the private sector (Faber, 2014). Its construction was mostly undertaken by local state-owned enterprises.[2] In fact, in each year during 1993–2007, about 80 counties on average were newly connected to the 5V7H, and the number of entrances/exits from counties to the 5V7H kept increasing year on year (Tian et al., 2024).

3. Conceptual framework

This section outlines a conceptual framework for understanding the potential impact of road construction on dietary quality in rural areas. Improvements in transportation infrastructure can affect dietary outcomes through both demand-side and supply-side channels.

On the demand side, road construction can help increase household income and improve dietary literacy, thereby improving dietary quality among rural residents. The demand for healthy foods depends on both adequate and stable income (Dercon et al., 2009; Huang and Tian, 2019; Ji et al., 2012; Mu and Van De Walle, 2011; Rammelt and Leung, 2017) and good dietary literacy (Zhao and Yu, 2019). However, rural residents in developing countries often have limited employment opportunities, and poor transportation infrastructure further constrains their ability to migrate for work (Asher and Novosad, 2020). In addition, traditional dietary norms and limited exposure to modern nutritional guidelines may reinforce unhealthy consumption habits (Min et al., 2021; Ren et al., 2019). Road construction can help alleviate these constraints by creating new local employment opportunities and reducing migration costs. Moreover, better roads may foster the diffusion of nutritional knowledge and increase exposure to healthier dietary norms through enhanced labor mobility. Together, these mechanisms can stimulate greater demand for healthy foods and ultimately improve dietary quality.

On the supply side, road construction can enhance dietary quality among rural residents by increasing food availability in rural areas. Rural communities in developing countries often face inadequate food supply, reflected in both limited variety and insufficient quantity (Remans et al., 2015). This is partly due to the lower purchasing power of rural households relative to their urban counterparts, which incentivizes suppliers to prioritize healthy foods for urban communities—leaving rural areas more vulnerable to "food deserts" (Bitler and Haider, 2010; Lewis et al., 2005; Powell et al., 2007). In addition, the dispersed nature of rural settlements increases the distance to markets, further limiting access to food (Asher and Novosad, 2020). By lowering transportation costs, road construction facilitates the delivery of a broader variety and larger quantities of food to rural markets. This process fosters market integration, ultimately making a diverse range of food both accessible and affordable for rural households. Simultaneously, improved infrastructure lowers travel costs for rural households, improving their access to food markets and expanding their dietary choices.

In sum, while road construction is expected to improve dietary quality among rural residents, the magnitude of this effect and the extent to which it is driven by demand-side or supply-side mechanisms remain empirical questions.

4. Data

To identify the impact of the 5V7H project on rural residents' dietary quality, we draw on data from two sources. One is the administrative geospatial data to measure rural residents' connection to the 5V7H system. The other is rural residents' dietary quality and socioeconomic status data from the China Health and Nutrition Survey (CHNS).

4.1. Connection to the 5V7H

To measure rural residents' connection to the 5V7H, we follow Liu et al. (2025) and take a five-step approach as follows. Firstly, we used the 1:1000,000 traffic map released by the National Bureau of Surveying and Mapping of China (2007) as the base and updated it annually against the "National Traffic Atlas" published by the China Map Press in relevant years (1997, 2000, 2004, and 2006). In this way, we obtained the e-map containing the 5V7H highways. Secondly, we obtained the county codebook for the CHNS,[3] which contains 32 counties and 16 cities from eight provinces. As this study focuses on rural residents, we excluded the 16 cities. Thus, the 32 counties became our study sample counties, which we located on the e-map. Thirdly, we identified the geographic boundaries of each sample county and located all entrances and exits of the 5V7H expressway within a sample county with the help of ArcGIS 10.2. Fourthly, we followed He et al. (2020) and Zhang and Chen (2025) to define counties whose geographic boundaries are traversed by the 5V7H highway network and contain at least one 5V7H entrance or exit as the treatment group, and those without such access as the comparison group. By this definition, during the construction period of the 5V7H system, 3, 4, and 2 sample counties became the treatment group by the end of 1997, 2000, and 2004, respectively. As could be imagined, once a county became the treatment group, it

[2] *A decade of action in transport: an evaluation of World Bank assistance to the transport sector, 1995-2005 (English)*. Washington, DC: World Bank. http://documents.worldbank.org/curated/en/808261468139200365

[3] Ge, K. (1998). *Health and Nutrition Status of Residents in Eight Provinces of China* (Vol. 1). Beijing: Beijing Science and Technology Press. [County codebook for the China Health and Nutrition Survey (CHNS).]

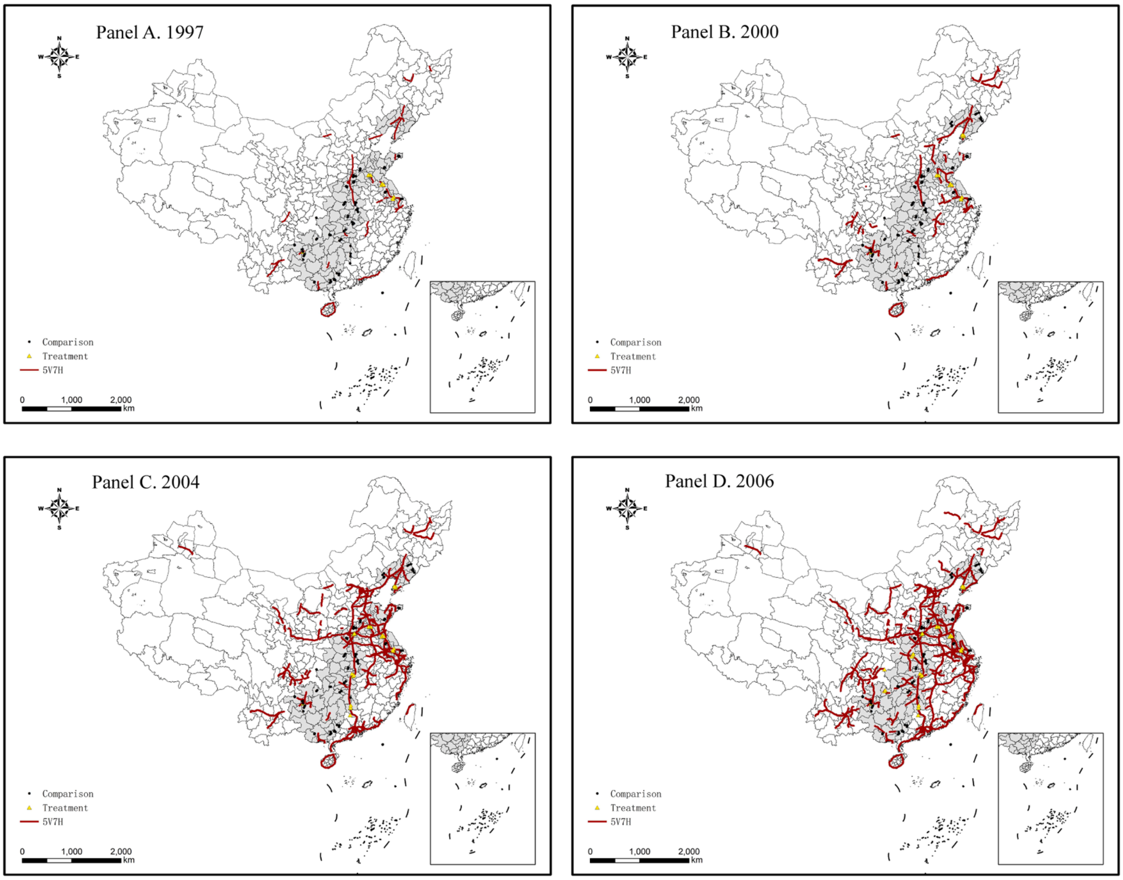

Fig. 1. Staggered rollout of the 5V7H between 1997 and 2006.
Notes: The yellow triangles and black dots denote the treatment group and comparison group, respectively. The red lines denote the 5V7H.

stayed in the treatment group for the rest of the study period. Therefore, by the end of 2007, 9 out of the 32 sample counties have become the treatment group and the rest 23 sample counties remain in the comparison group. In the final step, we considered all sample villages/households in the treatment (comparison) counties as treatment (comparison) villages/households. Members residing in the same household share the same treatment status as their household.

We use county rather than village as the unit of treatment/comparison classifications when studying the impact of the 5V7H connection on rural residents' dietary quality for two reasons. First, by the time the 5V7H was completed in China in 2007, a county along the 5V7H usually had a couple of highway entrances and exits. Second, when a county is connected to the outside market through the 5V7H highway, most of the commodities (including food) from outside of the county tend to be transported into the county center through the 5V7H highway first before they are distributed to places throughout the county by local vendors. Fig. 1 shows the distribution of sample counties by treatment status and by year. In addition, we also replace county with village as treatment/comparison classification unit as a robustness check, and the results remain substantially the same in both direction and magnitude.

4.2. China health and nutrition survey (CHNS)

We draw on information from the China Health and Nutrition Survey (CHNS) to measure the dietary quality and socio-economic status of rural residents. CHNS is jointly conducted by the Carolina Population Center at the University of North Carolina at Chapel Hill and the National Institute for Nutrition and Health at the Chinese Center for Disease Control and Prevention. It is a household-based cohort survey started in 1989, with approximately 4400 households interviewed each wave in urban and rural areas across nine provinces around China.[4] The samples were selected using a multi-stage randomized clustering strategy and are thought to be representative of the Chinese population (Zhang et al., 2013). Since its first survey wave conducted in 1989, eight follow-up survey waves have been conducted by 2011.[5]

[4] The nine provinces are Guangxi, Guizhou, Henan, Heilongjiang, Hubei, Hunan, Jiangsu, Liaoning, and Shandong.
[5] More information about the CHNS can be found at http://www.cpc.unc.edu/projects/china.

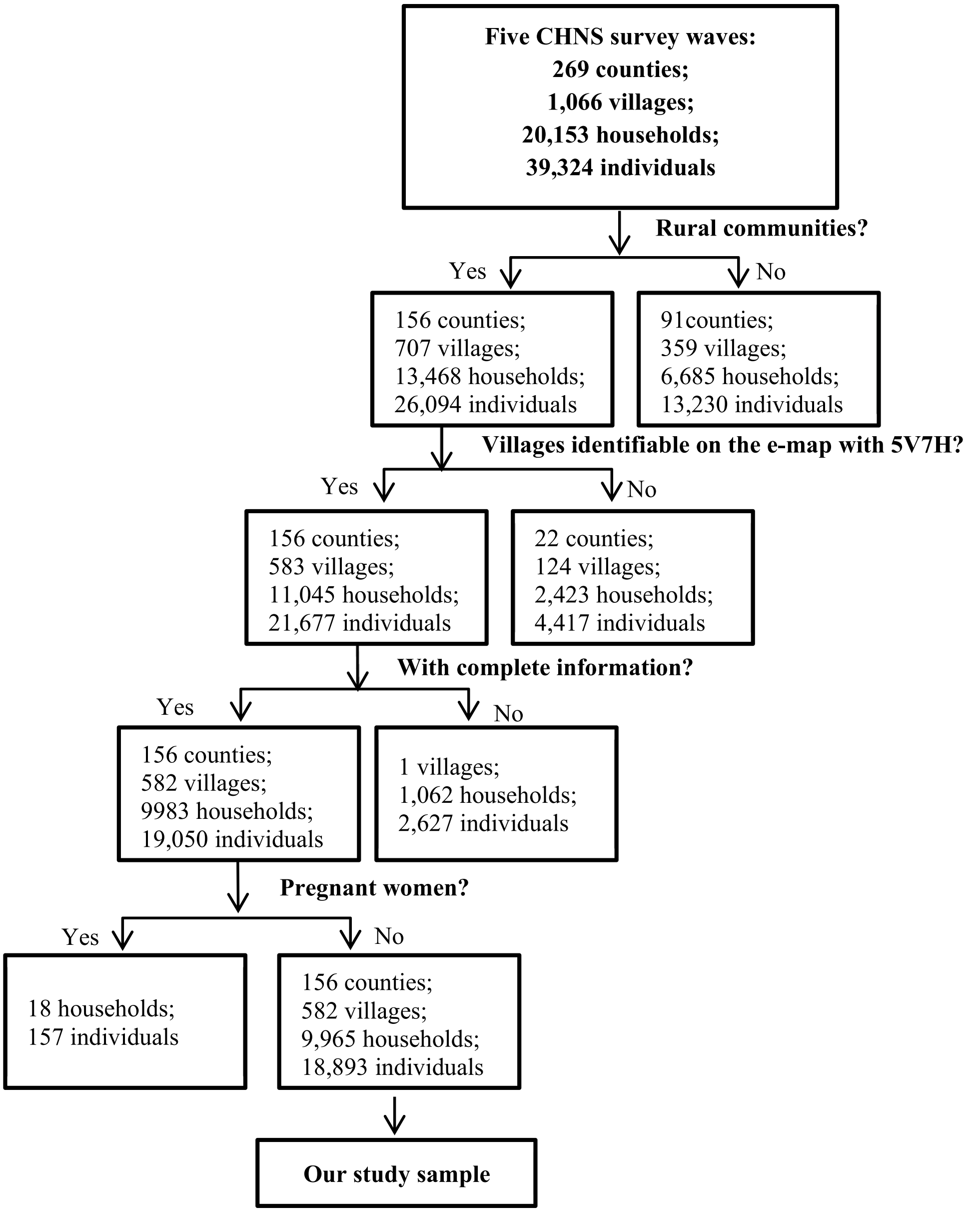

Fig. 2. Study sample construction from the CHNS.

Each CHNS wave collected rich information on the diet and socioeconomic status of households and household members. In addition, CHNS also collected detailed information at the village level, including village infrastructure (e.g., markets and stores, roads, sanitation facilities, etc.), population, wage rate, and food prices. These data will be used in the rest of this study.[6]

4.2.1. Construction of study sample

We take a seven-step procedure to merge the 5V7H connection data with the CHNS data to construct our study sample (Fig. 2). First, of the 1066 villages surveyed in these five CHNS waves (1997, 2000, 2004, 2006, and 2009), we excluded all those villages located in urban areas, which left us with 707 villages. Second, we merged these 707 villages with the 824 villages listed in the *Health and Nutrition Status of the Population in Eight Chinese Provinces*,[7] which left us with 583 villages. Third, we further removed villages

[6] The data used in this study were publicly available from CHNS. We also got the permission to use its community data. This study has been approved by the Ethics Committee of Georg-August-Universität Göttingen, the affiliation of one of the coauthors.

[7] Ge, K. (1998). *Health and Nutrition Status of Residents in Eight Provinces of China* (Vol. 1). Beijing: Beijing Science and Technology Press. [County codebook for the China Health and Nutrition Survey (CHNS).]

surveyed for only one year, which left us with 582 villages. Fourth, for these 582 villages, we obtained their latitude and longitude information from the HNSP8CP. Fifth, we located these 582 villages on the e-map containing the 5V7H highways. Sixth, for all those households and individuals that the CHNS surveyed in these 582 villages, we removed those with incomplete information, which left us with 9983 households and 19,050 individuals. Finally, considering the dietary restrictions on women during pregnancy and lactation (Gittelsohn et al., 1997), we followed Huang et al. (2022) and excluded those pregnant or lactating women. After these seven steps, we are left with 18,893 individuals from 9965 households at 582 villages in 32 counties of eight provinces, which become the study sample in the rest of the paper.

4.3. Measurements of rural residents' dietary quality

To measure rural residents' dietary quality, we drew on the detailed 24-hour food intake information for the past three consecutive days prior to the CHNS survey date for each household member. For the 1997, 2000, 2004, 2006, and 2009 CHNS survey waves,[8] we were able to get the food intake coding dictionary from the Chinese Food Composition Table (CFCT). This dictionary allows us to understand not only the food items that each household member consumed over the past three consecutive days, but also the nutrition facts per 100 *g* of food for each food type listed in the coding system, including water, carbohydrates, fat, protein, dietary fiber, cholesterol, fatty acids, and trace elements such as sodium, calcium, and zinc. Combining these two sources of information, we are able to construct indicators of dietary quality for each household member. As the construction period of the 5V7H spans between 1993 and 2007, and we are able to get the diet coding dictionary only for the 1997 CHNS survey wave and afterwards, we will use the data from the five CHNS survey waves conducted in 1997, 2000, 2004, 2006 and 2009 for the purpose of our study.[9]

We followed the literature and constructed two indicators to measure the dietary quality of rural residents: dietary diversity score (DDS) (Sibhatu et al., 2015; Snapp and Fisher, 2014), and healthy eating index (HEI) (Hu et al., 2022; Krebs-Smith et al., 2018). According to the Guidelines for Measuring Household and Individual Dietary Diversity provided by the Food and Agriculture Organization,[10] residents' dietary diversity could be assessed using DDS based on nine food groups as follows, starchy staples, dark green leafy vegetables, other vitamin-rich fruits and vegetables, other fruits and vegetables, organ meat, meat and fish, eggs, legumes, nuts and seeds, and milk and milk products. Detailed food group classifications and food items in each group can be found in Masters et al. (2018), and Sibhatu and Qaim (2018a), among others. The DDS was calculated by counting the number of food groups that an individual consumed in the past three consecutive days prior to the survey date without considering a minimum quantity requirement for any food group. Any individual food item in each food group consumed by an individual earns one point for his/her dietary diversity score, but different individual food items consumed in the same group are not counted repeatedly. Therefore, the DDS ranges from 0 to 9. The higher the DDS, the more diversified a resident's diet, and the better his/her dietary quality.

DDS is mainly about food diversity, we still need a measure of dietary health. Healthy Eating Index (HEI), an index originally introduced by (Kennedy et al., 1995), has been widely used to measure the dietary health of Americans (Guenther et al., 2013; Krebs-Smith et al., 2018). To measure the dietary health of Chinese residents, Yuan et al. (2017) developed Chinese HEI (CHEI) following the Dietary Guidelines for Chinese (DGC-2016) food intake. CHEI has been widely accepted and used in the context of China (Liu and Grunert, 2020; Wang et al., 2021; Zhao et al., 2021).

To construct CHEI for each resident, we follow Yuan et al. (2017) and take a four-step approach. Firstly, for each of the 17 food groups listed in Yuan et al. (2017),[11] we calculated the quantity (in grams) that a resident consumed in the past three consecutive days prior to the survey date and divided it by three to get its daily consumption. Secondly, for each of the 17 food groups, we converted the daily consumption to a standard intake of 1000 calories (Standard Portion/1000 kcal) according to the Chinese Dietary Nutrition Guidelines (2016). Thirdly, following the scoring manual of Yuan et al. (2017), we scored each of the 17 food groups based on their standard portion intake. Finally, we summed the scores for each of the 17 food groups to obtain CHEI. Thus, the CHEI ranges between 0 and 100 points. The higher the index, the better his/her dietary quality.

5. Empirical strategy

The key to rigorously identifying the causal effects of the 5V7H expressway on rural residents' dietary quality lies in the exogeneity of their connection to the highway network. Without addressing this issue, estimated effects may be biased by endogenous infrastructure placement. Many studies have examined the economic consequences of China's national expressway system, covering a wide

[8] As such information is not available for the 1989, 1991, and 1993 survey waves of the CHNS, we did not include these earlier three survey waves in this study.

[9] As the diet coding dictionary for the 1991 and 1993 survey waves was not available from the Chinese Food Composition Table, we did not include these two waves in this study. We also had to forgo the survey waves after 2009 to avoid the confounding of other national trunk systems or high-speed train railways built after 2007. In short, this study drew on data from the five survey waves between 1997 and 2009 of the CHNS.

[10] Kennedy, G., Ballard, T., & Dop, M. C. (2011). *Guidelines for measuring household and individual dietary diversity*. Rome: Food and Agriculture Organization of the United Nations (FAO). https://www.fao.org/fileadmin/user_upload/wa_workshop/docs/FAO-guidelines-dietary-diversity2011.pdf

[11] Of the 17 food groups, 12 groups (including total grains, whole grains and mixed beans, tubers, total vegetables, dark vegetables, fruits, dairy, soybeans, fish and seafood, poultry, seeds and nuts, and eggs) whereas the rest 5 groups (including red meat, cooking oils, sodium, added sugar, and alcohol) are negatively scored. In this way, higher CHEI means healthier diet.

range of outcomes such as economic growth and environmental change (He et al., 2020), the diffusion of industrial activity (Faber, 2014), allocative efficiency (Wu et al., 2023), industrial clustering (Zhang and Chen, 2025), poverty alleviation (Emran and Hou, 2013; Tian et al., 2024), income inequality (Lu et al., 2022), and labor market dynamics (Falaris and Yang, 2025). Among them, Faber (2014) and He et al. (2020) are particularly relevant to our study as both explicitly address the identification challenge associated with the non-random placement of China's national highway network. Both studies note that since China's national highway system was originally designed to connect targeted nodal cities, the connections to those cities cannot be treated as exogenous. They therefore focus on non-nodal counties that were not explicitly targeted by the original network plan but were connected during the course of network expansion, and identify causal effects within this selected sample using IV and DID approaches, respectively.

Following Faber (2014) and He et al. (2020), we focus on rural counties rather than targeted nodal cities and employ a county-level staggered difference-in-differences (DID) design to identify the effects of 5V7H connection on rural residents' dietary quality. The treatment group consisted of all individuals living in counties whose geographic boundaries were traversed by the 5V7H highway network and had at least one 5V7H entrance/exit during the study period. The comparison group included all individuals from counties whose geographic boundaries did not have any 5V7H entrance/exit during the study period. Following Tian et al. (2024), we estimate the following empirical specification,

$$y_{jit} = \alpha + \beta Connect_{jt} + \gamma Z_{jit} + \mu_j + \theta_m + \rho_t + \varepsilon_{jit} \tag{1}$$

here subscripts j, i and t denote county, individual and year, respectively. y_{jit} is the dietary quality (DDS or CHEI) in year t for individual i from county j. $Connect_{jt}$ indicates whether county j was connected to the 5V7H system in year t, which takes the value of one if connected and zero otherwise. Z_{jit} is a vector of control variables at the individual/household/county levels that might affect rural residents' dietary quality. Following the literature, we control for six individual and household-level variables: gender dummy, age in years, ethnic minority dummy, years of education, a dummy indicating whether the individual experienced any fever, stomachache, headache, or heart attack in the four weeks preceding the CHNS survey, and household size (Vemireddy and Pingali, 2021). In addition, we incorporate province-year and county-month fixed effects in the robustness checks to account for time-varying policy shocks and seasonal factors.

There is concern that the impact of the 5V7H connection may be driven by pre-existing trends. To address this concern, we used an event study approach to examine whether rural residents with connection to the 5V7H were significantly different from rural residents without connection to the 5V7H in the early years. Specifically, we estimated the following equation:

$$y_{jit} = \sum_{k \geq -4,\ k \neq -1}^{k=4} D_{jt}^{k} \cdot \delta_k + \gamma Z_{jit} + \mu_j + \theta_m + \rho_t + \gamma_{jt} + \varepsilon_{jit} \tag{2}$$

here D_{jt}^{k} is a dummy variable defined as follows: $D_{jt}^{k} = 0$ for any k and t for county j that never or always had a connection to the 5V7H during the sample period. For county j that had a connection to the 5V7H within the sample period, we first define ρ_j as the year when the county was first connected to the 5V7H, and then define $D_{jt}^{k} = 1$ if $t - \rho_j = k$, and zero otherwise, where $k \in \{-3, -2, 0, 1, 2, 3\}$. Note that the post-treatment effects are relative to the year before the 5V7H connection as Eq. (2) omits the $k = -1$ dummy variable. The other variables are the same as described in Eq. (1).

Another concern is that the 5V7H planners might have planned arbitrarily those countries with development potential into the 5V7H road network. To mitigate this endogeneity concern, we combine DID and instrumental variable (IV) approaches by adapting the strategy proposed by Faber (2014) to our study context. To do so, we need to construct the Euclidean spanning tree network with the 48 cities along the 5V7H as connection nodes, and use these road dendritic networks as IV for the 5V7H connections. The validity of such an IV can be justified as follows. First, constructing roads between major cities with the spanning tree network is the most desirable construction scenario (Faber, 2014), which is not likely to be affected by the counties' development potential or political status. Therefore, a county that happens to be in the vicinity of a virtual road tree-like network can be viewed as random. Second, such an IV cannot affect rural residents' dietary quality unless through their connections to the 5V7H.

However, the virtual Euclidean spanning-tree networks that we constructed by following Faber's (2014) strategy may not fit our staggered DID model as it is not staggered. Therefore, we consider the canonical difference-in-differences strategy (two periods and two groups) and write out the DID estimation specification as follows:

$$y_{jit} = \alpha + \beta Treatment_j * Post_t + \gamma Z_{jit} + \mu_j + \theta_m + \rho_t + \varepsilon_{jit} \tag{3}$$

where $Treatment_j$ indicates whether county j was connected to the 5V7H system during 1997–2006, which takes the value of one if connected and zero otherwise. $Post_t$ denotes an indicator variable that equals 1 for the periods after 2004, as 89% of the 5V7H connections had been completed by the end of 2003 (Faber, 2014). The other variables are the same as described in Eq. (1). Then for each sample county, we measured the distance from its center to the virtual Euclidean spanning-tree, which we as the IV for the 5V7H connection.

Still another concern with the staggered DID was estimation bias due to potential heterogeneity in treatment effects. As the date of connection to the 5V7H varies by county, the estimated coefficient of a given lead or lag might be contaminated by effects from other periods (Sun and Abraham, 2021), resulting in a biased estimate of treatment effects when using staggered DID. To deal with this concern, we followed Baker et al. (2022) to use the estimators developed by Callaway and Sant'Anna (2021) and Sun and Abraham (2021), respectively.

Table 1
Summary statistics.

Variable	Mean	SD
Dietary Quality		
DDS-9	4.62	1.29
CHEI	50.41	8.12
Connection to 5V7H		
5V7H connection (1= Yes, 0= No)	0.20	0.40
Individual and Household Characteristics		
Female (1= Yes, 0= No)	0.51	0.50
Age (Years)	46.58	16.11
Ethnic Minority (1= Yes, 0= No)	0.16	0.37
Years of education	6.24	4.16
Ever sick over the past 4 weeks (1= yes, 0= No)	0.11	0.32
Household size (Person)	3.81	1.48
Regional Characteristics		
New connections to other highways constructed after 2007 (1= Yes, 0= No)	0.02	0.13

Notes: Data on dietary quality and individual and household characteristics are drawn from the 1997, 2000, 2004, 2006, and 2009 waves of the China Health and Nutrition Survey (CHNS). Data on the 5V7H connection and new connections to other highways constructed after 2007 are based on authors' calculations.

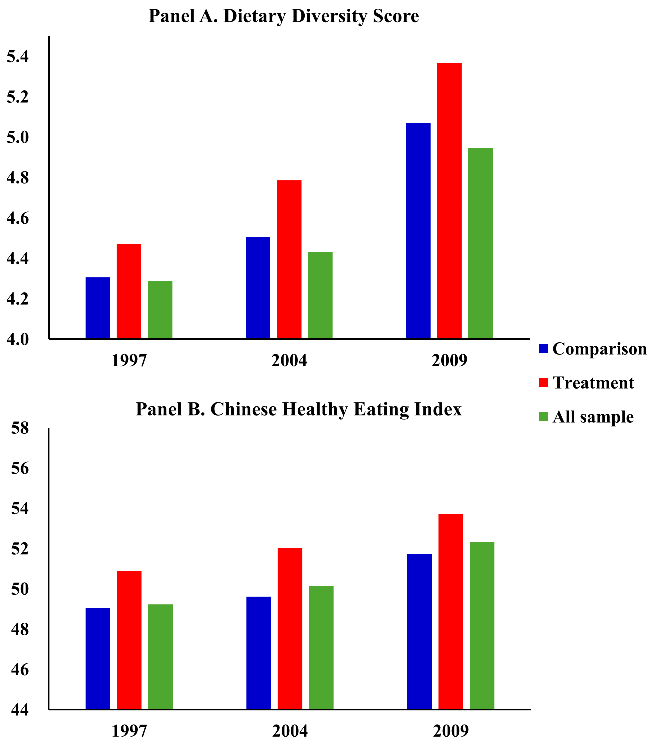

Fig. 3. Summary statistics of dietary quality, by Year and treatment status.

6. Empirical results

6.1. Descriptive statistics

Fig. 1 presents the staggered rollout of the 5V7H. Our data show that 20 percent of the sample counties were successively connected to the 5V7H between 1997 and 2009 (Table 1). Of the sample rural residents, their mean DDS and CHEI were 4.62 and 50.41, respectively; 51 percent were female, 16 percent were ethnic minorities, and 11 percent self-reported ever being sick in the past four weeks prior to the CHNS survey date. An average rural resident was 46.58 years old, had 6.24 years of education, and came from a household with 3.81 members.

Descriptive statistics show the dietary quality of rural residents during the study period has been improving, especially for those with connection to the 5V7H (Fig. 3). Between 2000 and 2009, the DDS (CHEI) increased from 4.46 (48.61) to 5.07 (52.32) for the

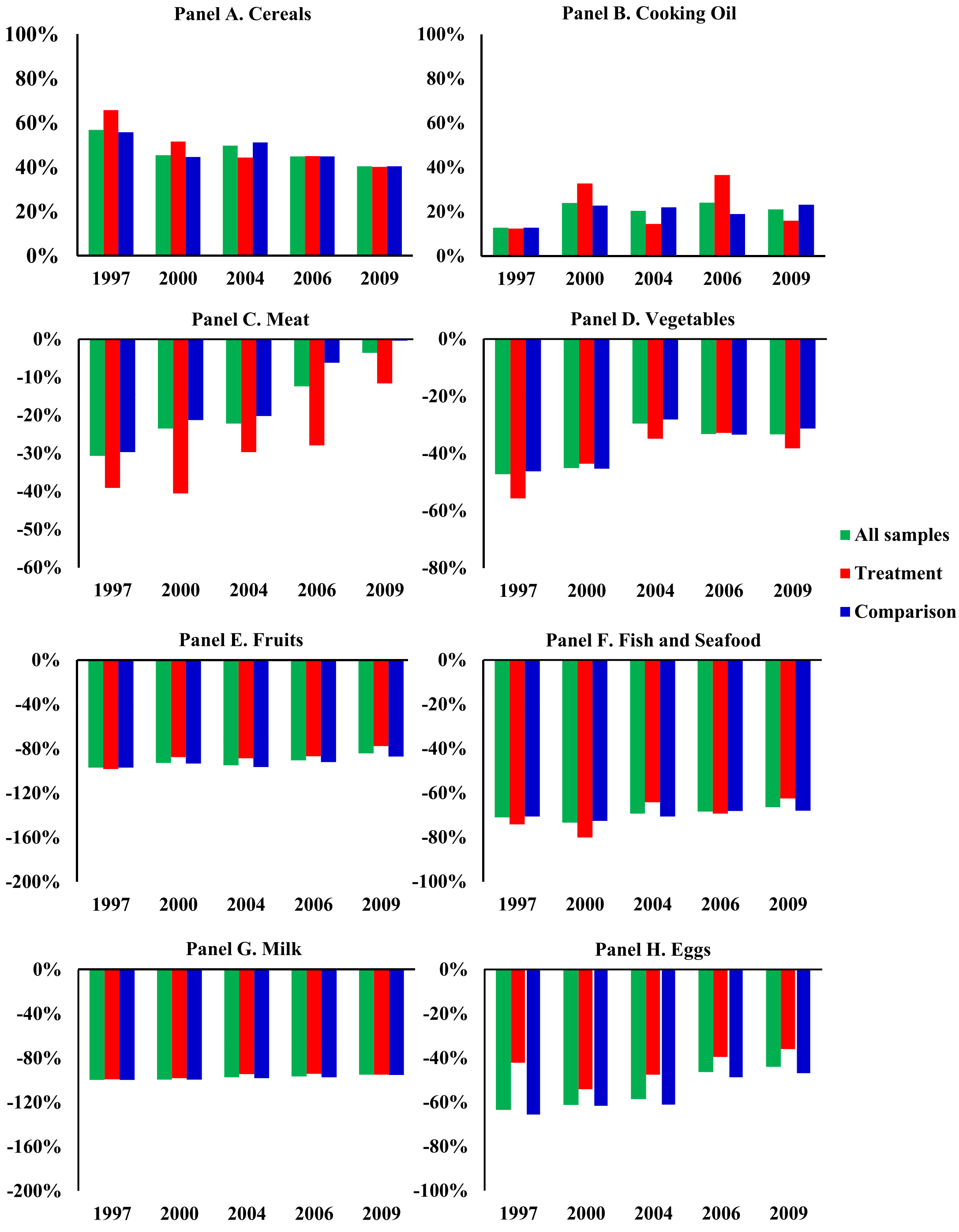

Fig. 4. Deviations of food intake from values recommended by the Dietary Guidelines, by Year and treatment status.
Note: The deviation is calculated as (food intake - recommended diet) / recommended diet * 100%.

study sample as a whole, with more increase among those with connection to the 5V7H (from 4.49 to 5.37 for DDS and from 48.67 to 53.72 for CHEI) than those without (from 4.46 to 4.95 for DDS and 48.60 to 51.74 for CHEI). In other words, the treatment group (those with connection) experienced 0.39 (1.90) more increase in mean DDS (CHEI) than the comparison group (those without connection). This provides suggestive evidence that the 5V7H connection is positively correlated with the dietary quality of rural residents (Panel A).

We find additional descriptive evidence for the positive correlation between the 5V7H connection and rural residents' dietary quality when examining the variation of food intake from values recommended by the Dietary Guidelines (Fig. 4). For example, the

Table 2
Impacts of 5V7H connection on rural residents' dietary quality.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dietary Diversity Score (DDS)				Chinese Healthy Eating Index (CHEI)			
5V7H Connection	0.363***	0.326***	0.241**	0.461***	2.235***	2.197***	2.111**	2.377**
	(0.107)	(0.108)	(0.106)	(0.133)	(0.772)	(0.785)	(0.855)	(1.132)
	(0.114)	(0.117)	(0.133)	(0.157)	(1.128)	(1.151)	(1.330)	(1.710)
	(0.102)	(0.106)	(0.119)	(0.136)	(0.757)	(0.791)	(0.976)	(1.147)
Individual and household characteristics	no	yes	yes	yes	no	yes	yes	yes
Regional characteristics	no	yes	yes	yes	no	yes	yes	yes
Provincial Trends	yes	yes	no	yes	yes	yes	no	yes
County FE	yes	yes	yes	no	yes	yes	yes	no
Survey Month FE	yes	yes	yes	no	yes	yes	yes	no
Year FE	yes	yes	no	yes	yes	yes	no	yes
Province- Year FE	no	no	yes	no	no	no	yes	no
County- Survey Month FE	no	no	no	yes	no	no	no	yes
Observations	18,893	18,893	18,893	18,889	18,893	18,893	18,893	18,889
Adjusted R-squared	0.293	0.318	0.329	0.345	0.138	0.169	0.179	0.187

Notes: Control variables include individual and family characteristics (gender, age, nationality, years of education, sick and household size), and regional characteristics (new connections after 2007).
We test the robustness of the estimation precision by clustering standard errors at three different levels: county-year, county, and province-year. These standard errors are sequentially reported in parentheses below the estimated coefficients. Our preferred model specification uses standard errors clustered at the county-year level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All regressions are weighted by the province-level sample size.

over-intake of cereals against the recommendation decreased more among the treatment group than the comparison group (a decrease of 25.64 percent versus 15.28 percent). As for cooking oil, although the over-intake against the recommendation was almost the same for both the treatment and comparison groups in 1997, by 2009, the over-intake for the treatment group was much smaller than the comparison group (an over-intake of 15.93 percent versus 23.18 percent). When we look at the food groups that rural residents have inadequate intake, the magnitude of under-intake against the recommendation has been consistently smaller among the treatment group than the comparison group for fruits (87.82 percent versus 93.14 percent), milk (96.20 percent versus 98.00 percent), and eggs (43.88 percent versus 57.22 percent). However, the picture looks almost the opposite for meat and vegetables whose under-intake against the recommendation was more among the treatment group than the comparison group (29.76 percent versus 15.53 percent for meat and 41.02 percent versus 36.85 percent for vegetables).

6.2. Regression results

6.2.1. Results from the staggered DID approach

Regression results from the staggered DID consistently show that the 5V7H connection improved rural residents' dietary quality (Table 2). Compared to their rural peers without connection to the 5V7H, residents in connected counties reported, on average, 0.326 more food groups in DDS and scored 2.197 points higher in CHEI (Columns 2 and 6). Given that the average in the baseline is 4.62 for DDS 50.41 for CHEI (Table 1), these estimates imply increases of approximately 7.1 percent and 4.4 percent, respectively. In other words, these estimated effects imply that compared with their rural peers without the 5V7H connection, rural residents with the connection not only exhibit more dietary diversity, but also have diets that are more aligned with the recommended dietary pattern. Using the full CHEI score (100 points) as the benchmark for a diet fully consistent with the guidelines, the estimated effect is equivalent to a 4.4 percent reduction in the gap between the average CHEI score (50.41 points) at baseline and that benchmark.[12] Such magnitudes indicate a meaningful improvement in both dietary diversity and overall dietary quality associated with highway connection. The results remain robust when standard errors are clustered at different levels.

We next tested the robustness of the results to alternative fixed-effects specifications. Specifically, we replaced year fixed effects with province-year fixed effects to absorb time-varying unobserved factors at the provincial level. To further address potential seasonality across regions, we included county-month fixed effects to capture seasonal variations within counties over time. The results remain robust, confirming that the 5V7H connection significantly improved rural residents' dietary quality (Columns 3–4 and 7–8).

Furthermore, to better understand which food groups drive these overall improvements, we examine the impact of the 5V7H connection on the components of CHEI (Table 3). Compared to their rural peers without 5V7H connection, rural residents with such connection scored significantly higher in the CHEI specifically for fruits (0.935), fish and seafood (0.633), nuts (0.264), milk (0.165), and oil (0.656), while scoring lower for sugar (–0.018), and sodium (–0.459). These findings suggest that the 5V7H connection promoted the intake of recommended food groups such as fruits, aquatic products, nuts, and milk, while reducing oil intake. However, it also increased the intake of condiments (as reflected by lower scores in sugar and sodium).

[12] The 4.4 percent figure is calculated as 2.197/ (100-50.41) = 4.43%, where 2.197 is the estimated CHEI effect, 50.41 is the sample mean of CHEI, and 100 is the full CHEI score. CHEI ranges from 0 to 100, with higher scores indicating better dietary quality and adherence to the Dietary Guidelines for Chinese (DGC-2016) at a given level of energy intake.

Table 3
Impact of 5V7H connection on the CHEI subproject scores.

	(1) Cereals	(2) Mix Beans	(3) Root	(4) Vegetables	(5) Dark vegetables	(6) fruits	(7) nut	(8) milk	(9) bean	(10) Fish and Seafood	(11) Wheat meat	(12) Red meat	(13) Egg	(14) Oil	(15) Sugar	(16) Alcohol	(17) Na+
5V7H Connection	0.059	−0.077	0.120	0.217	−0.188	0.935***	0.264***	0.165**	0.063	0.633***	0.041	0.057	−0.244	0.656**	−0.018*	−0.028	−0.459*
	(0.107)	(0.151)	(0.239)	(0.172)	(0.146)	(0.337)	(0.094)	(0.071)	(0.225)	(0.130)	(0.122)	(0.089)	(0.161)	(0.289)	(0.010)	(0.022)	(0.265)
Individual and household characteristics	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Regional characteristics	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Provincial Trends	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
County FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Survey month FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893	18,893
R-squared	0.100	0.133	0.129	0.127	0.267	0.205	0.089	0.066	0.118	0.282	0.179	0.290	0.158	0.090	0.011	0.063	0.090

Notes: Control variables include individual and household characteristics (gender, age, nationality, years of education, sick and household size), and regional characteristics (new connections after 2007).
Standard errors clustered at the county-year level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.
The regressions are weighted by the province-level sample size.

Table 4
Impacts of 5V7H connection on rural residents' BMI, overweight, and obesity.

	(1) BMI	(2) Overweight	(3) Obesity
5V7H Connection	−0.030 (0.106)	−0.006 (0.015)	−0.010** (0.004)
Individual and household characteristics	yes	yes	yes
Regional characteristics	yes	yes	yes
Provincial Trends	yes	yes	yes
County FE	yes	yes	yes
Year FE	yes	yes	yes
Survey month FE	yes	yes	yes
Observations	17,070	17,070	17,070
R-squared (pseudo)	0.117	0.070	0.095

Notes: Control variables include individual and household characteristics (gender, age, nationality, years of education, sick and household size), and regional characteristics (new connections after 2007).
Standard errors clustered at the county-year level and reported in parentheses. * $p < 0.1$;.
** $p < 0.05$; *** $p < 0.01$.
Columns 2–3 report marginal effects based on probit estimation.

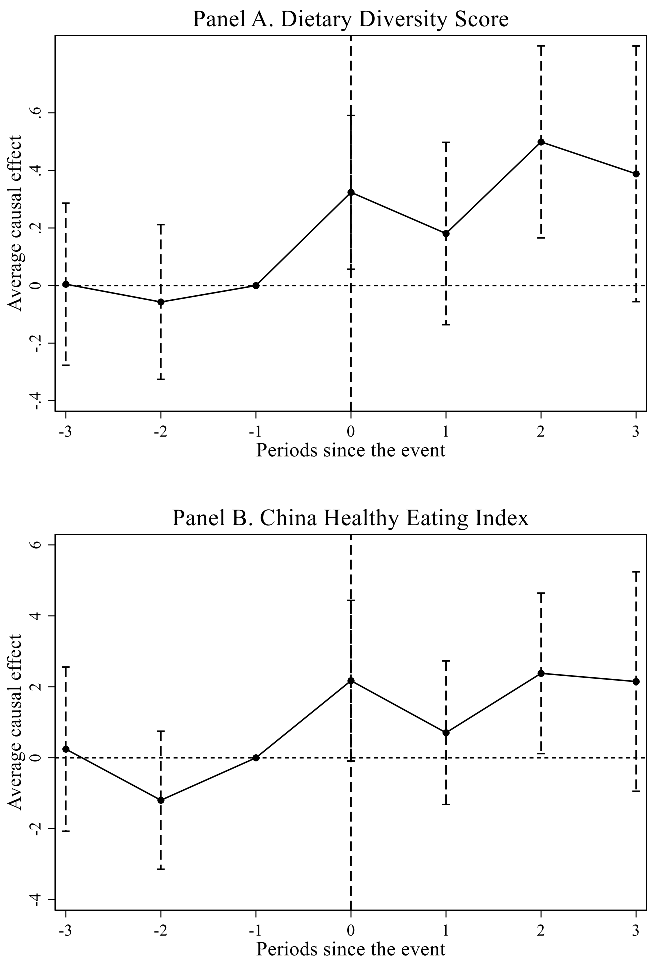

Fig. 5. Parallel trend tests.
Notes: The figure presents the estimates and their 95% confidence intervals from the event study. The X-axis indicates the years before or after the 5V7H connection, and the Y-axis indicates the estimated average treatment effect. The regressions are weighted by the province-level sample size.

Table 5
Robustness checks.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Alternative Estimation Strategies						Control for Confounding Factors			Account for potential migration effects	
	Village Identification Strategy		Canonical DID (TWFE)	Canonical DID (IV)	SA Estimator	CS Estimator	Distance to railway	No. of household vehicles	Both		
	Staggered DID	Log distance									
Panel A. Dietary Diversity Score (DDS)											
5V7H Connection	0.262**	−0.065***	0.294**	0.532**	0.310***	0.394**	0.343***	0.317***	0.331***	0.323***	0.338***
	(0.102)	(0.022)	(0.130)	(0.232)	(0.086)	(0.162)	(0.108)	(0.108)	(0.108)	(0.112)	(0.109)
5V7H Connection × household with migrant members (1=Yes; 0=No)											−0.052 (0.074)
Kleibergen-Paap Wald F				30.372							
Observations	18,893	18,893	18,893	18,893	18,893	16,852	18,893	18,893	18,893	15,703	18,893
R^2	0.384	0.317	0.318	0.040	0.299	/	0.320	0.325	0.327	0.312	0.318
Panel B. Chinese Healthy Eating Index (CHEI)											
5V7H Connection	2.531***	−0.397**	1.758**	5.786***	2.224***	2.145*	2.277***	2.141***	2.210***	2.252***	2.126***
	(0.727)	(0.193)	(0.843)	(1.668)	(0.429)	(0.993)	(0.803)	(0.765)	(0.780)	(0.823)	(0.801)
5V7H Connection × household with migrant members (1=Yes; 0=No)											0.288 (0.423)
Kleibergen-Paap Wald F				30.372							
Observations	18,893	18,893	18,893	18,893	18,893	16,852	18,893	18,893	18,893	15,703	18,893
R^2	0.196	0.168	0.169	0.033	0.145	/	0.170	0.175	0.176	0.164	0.169

Notes: Panel A and Panel B report results using the Dietary Diversity Score (DDS) and the Chinese Healthy Eating Index (CHEI) as dietary quality indicators, respectively.
Column 1 uses a village-level identification strategy, where villages within 10 km of the nearest 5V7H entrance/exit are defined as treated. Column 2 replaces the 5V7H connection indicator with the straight-line distance from the county centroid to the 5V7H network. Column 3 adopts a simplified canonical DID model with only two periods (pre- and post-2004). Column 4 follows Faber (2014) and uses the distance from the village to the virtual Euclidean spanning tree as the IV for the 5V7H connection, based on the simplified canonical DID model, and additionally controls for the interaction between distance to the nodal city and year. Columns 5–6 follow Sun and Abraham (2021) and Callaway and Sant'Anna (2021) and re-estimate the results. Columns 7–9 control for confounding factors, including railway, household vehicles, and secondary roads, respectively. Columns 10–11 account for potential migration effects. Column 10 restricts the sample to households without migrant members, and Column 11 includes an interaction term between the 5V7H connection and the migrant-household dummy.
Unless otherwise noted in the specific columns above, all regressions control for the individual/household/regional characteristics as discussed above, provincial trends, year fixed effects, county fixed effects, and survey month fixed effects; standard errors are clustered at the county-year level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All regressions are weighted by the province-level sample size.

Table 6
The impacts of rural road connection.

	(1) DDS	(2) CHEI	(3) DDS	(4) CHEI	(5) DDS	(6) CHEI	(7) DDS	(8) CHEI
5V7H Connection					0.338***	2.298***	0.174	1.181
					(0.113)	(0.804)	(0.126)	(0.914)
Village with paved road	−0.004	0.039	0.491**	3.735***	0.009	0.125	−0.022	−0.088
(1=Yes, 0=No)	(0.062)	(0.429)	(0.219)	(1.353)	(0.061)	(0.424)	(0.070)	(0.478)
Interaction term							0.200*	1.361*
							(0.107)	(0.707)
Individual and household characteristics	yes	yes	yes	yes	yes	yes	yes	yes
Regional characteristics	yes	yes	yes	yes	yes	yes	yes	yes
Provincial Trends	yes	yes	yes	yes	yes	yes	yes	yes
Village FE	yes	yes	yes	yes	yes	yes	yes	yes
Survey Month FE	yes	yes	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes	yes	yes
Observations	18,893	18,893	1596	1596	18,893	18,893	18,893	18,893
R-squared	0.388	0.198	0.301	0.227	0.390	0.200	0.390	0.201

Notes: Control variables include individual and household characteristics (gender, age, nationality, years of education, sick and household size), and regional characteristics (new connections after 2007).
Columns (3) – (4) use a restricted sample of villages where main road was not paved until after 2004 (N = 1596). Columns (1) – (2) and (5) – (8) use the full sample (N = 18,893).
Standard errors clustered at the village level and reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Finally, we extend the analysis to residents' health outcomes in terms of Body Mass Index (BMI) and the prevalence of overweight and obesity (Table 4). The results show that the 5V7H connection significantly reduced obesity prevalence among rural residents. Although its estimated effects on BMI and overweight status are also negative, they are not statistically significant. These findings suggest that improved transportation infrastructure contributes to not only better dietary quality, but also better health outcomes among rural residents.

The validity of the regression results from the staggered DID lies in the parallel trends. Fig. 5 presents the estimated coefficients of the leads and lags as well as their 95 percent confidence intervals from the event study as specified in Eq. (2). Both panels show that rural residents in the treatment group (with 5V7H connection) had a dietary quality similar, in terms of both DDS and CHEI, to that of rural residents in the comparison group (without 5V7H connection) before the 5V7H connection. However, after the 5V7H connection, the treatment group exhibited significantly better dietary quality than the comparison group. Therefore, our data provide strong support for the parallel trend assumption. We also follow Sun and Abraham (2021) to account for treatment effect heterogeneity and found no pre-treatment effects, lending additional evidence in support of the parallel trends assumption (Appendix Fig. 1). In contrast, post-treatment effects display a consistent upward trend, demonstrating the robustness of our baseline findings.

Moreover, we test for pre-existing differential trends in key covariates used in Eq. (1). Specifically, we use a standard event-study framework (Eq. (2)), with the period immediately before the 5V7H connection ($t = -1$) as the reference category. For each non-outcome covariate, we examine differences between connected and non-connected counties in both pre- and post-connection periods, focusing particularly on the two periods prior to treatment ($t = -2$ and $t = -3$). Across all covariates, we find little differences in the pre-treatment periods, indicating no diverging trends in observable characteristics before the 5V7H rollout (Appendix Table 1). These results lend further support to the parallel- trend assumption and strengthen the credibility of our staggered DID design.

6.3. Robustness checks

We conducted a series of robustness checks. Table 5 presents the results in two panels: one for DDS and the other for CHEI. Overall, our results are quite robust.

Results from the Village Identification Strategy: We define those villages with the shortest straight-line distance <10 km from the nearest 5V7H highway entrance/exit as the treatment group and the rest as the comparison group, and rerun Eq. (1). Consistent with the county-level staggered DID results, results from the village-level staggered DID results confirm that proximity to the 5V7H highway is associated with improved dietary quality (Column 1). We also find that shorter distances from the village committee to the highway are consistently associated with better dietary outcomes (Column 2). Specifically, a one standard deviation increase in the log-distance is associated with a reduction of approximately 0.090 points in DDS ($-0.065 \times 1.38 = -0.090$) and a decrease of about 0.538 points in CHEI ($-0.390 \times 1.38 = -0.538$).

Results from the canonical DID strategy: We define all survey waves before 2004 as pre-treatment period and the rest as post-treatment period to take a canonical DID strategy. Moreover, we follow Faber (2014) and use the distance from the county centroid to the virtual Euclidean spanning tree as the IV for the 5V7H connection, and additionally controls for the interaction between distance to the nodal city and year. Consistent with results from the staggered DID, regression estimates from both the canonical DID strategy and the IV combined with canonical DID also show that the 5V7H connection improved rural residents' dietary quality (Columns 3–4).

Results from the estimators by** Sun and Abraham (2021) **and** Callaway and Sant'Anna (2021)**: Both estimators confirm positive

Table 7
Exploration of underlying mechanisms: supply side.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
	Distance to Food Stores (km)			Food Price (kg/yuan)													
	Nearest	3-Nearest Avg.	5-Nearest Avg.	Premium rice	Ordinary rice	Premium wheat flour	Ordinary wheat flour	Rapeseed oil	Peanut oil	Green vegetables	Cabbage	Egg	Pork	Chicken	Beef	mutton	Carp
5V7H Connection	−2.625**	−3.359**	−4.377**	0.009	−0.017*	−0.216*	−0.079*	−0.064	0.031	−0.109**	−0.023	0.133**	−0.090	−0.108	−0.146	−0.225	−0.048
	(1.029)	(1.567)	(1.933)	(0.021)	(0.009)	(0.118)	(0.042)	(0.041)	(0.068)	(0.055)	(0.026)	(0.064)	(0.069)	(0.122)	(0.110)	(0.193)	(0.057)
Regional characteristics	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Provincial Trends Provincial Trends	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
County FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Survey Month FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	582	582	582	548	549	390	505	376	375	525	565	568	573	469	544	484	534
R^2	0.338	0.373	0.388	0.443	0.530	0.246	0.216	0.476	0.546	0.195	0.233	0.364	0.468	0.664	0.539	0.534	0.548

Notes: Control variables include regional characteristics (new connections after 2007).
Standard errors clustered at the county-year level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All food prices are deflated using the 1991-based constant food price index to ensure comparability over time.

Table 8
Exploration of underlying mechanisms: demand side.

	(1) Share of household labor off-farm (%)	(2) Per capita household income over the past year (10,000 CNY)	(3) Number of information tools (TV, mobile phone and computer)	(4) Total Dietary Knowledge Score	(5) Composite Dietary Knowledge Score	(6) Household cooking pattern: frying and stir-frying ratio (%)
5V7H Connection	4.745*	0.094*	0.147*	0.881***	0.204***	−4.110***
	(2.654)	(0.050)	(0.084)	(0.151)	(0.042)	(1.556)
Individual and household characteristics	yes	yes	yes	yes	yes	yes
Regional characteristics	yes	yes	yes	yes	yes	yes
Provincial Trends	yes	yes	yes	yes	yes	yes
County FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Survey month FE	yes	yes	yes	yes	yes	yes
Observations	18,893	18,887	18,893	10,221	10,221	18,893
R-squared	0.181	0.184	0.467	0.449	0.412	0.282

Notes: Control variables include individual and household characteristics (gender, age, nationality, years of education, sick and household size), and regional characteristics (new connections after 2007).
Per capita household income is deflated using the 1991-based constant food price index to ensure comparability over time.
Standard errors clustered at the county-year level and reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

effects of the 5V7H connection on the dietary quality of rural residents, although the magnitude of the estimated coefficients varies slightly (Columns 5–6).

Confounding factors: We consider two factors that might confound the 5V7H connections and dietary quality of rural residents. One is the connection to the railway (measured by the distance from the village committee to the nearest railway) as the construction of highways and railroads in China overlaps sometimes (Li and DaCosta, 2013; Lu et al., 2022). The other is the household's possession of transportation vehicles (measured by the number of cars and motorcycles possessed) as it affects the range of activities of rural residents (Huang and Tian, 2019; Kumar et al., 2015; Usman and Haile, 2022). Omitting these two factors may overestimate the impact of the 5V7H connection on rural residents' dietary quality, so we control for them separately and rerun Eq. (1). As shown in Columns 7–9, the regression results are robust to controlling for these two factors.

Potential migration bias: We further examine whether migration induced by the 5V7H connection may bias our estimation results. Based on the residence information provided by the CHNS, we construct a dummy variable of migrant households, which takes the value of one if at least one household member resides outside the county for at least three months in the survey year and zero otherwise. We re-estimate Eq. (1) using only the sub-sample of non-migrant households, the results are almost identical to those from the full sample (0.323 vs. 0.326 for DDS; 2.252 vs. 2.197 for CHEI). Then we introduce an interaction term between the 5V7H connection and the migrant-household dummy into Eq. (1) to capture potential heterogeneous effects and re-estimate with the entire study sample, the interaction term is statistically insignificant (Columns 10–11). These findings suggest that migration is not likely to bias our main conclusions.

Complementary relationship between the 5V7H connection and rural roads: There has been evidence that rural roads contributes to the welfare of rural populations (Aggarwal, 2021, 2018; Rammelt and Leung, 2017). We conducted two exercises to test whether there is any interaction between the impacts of the 5V7H connection and rural roads on the dietary quality of rural residents. First, we replace the 5V7H connection in Eq. (1) with a dummy variable that takes the value of one if the main road in the village is paved (concrete or asphalt), and zero otherwise and re-estimate it. As shown in Table 6, the estimated coefficient of paved village road is insignificant (Columns 1–2). One plausible explanation is 72% of sample villages had paved roads before 2004. In fact, when we reran the equation using only villages where main road was not paved until after 2004, we found paved village road exert significant positive effects on residents' dietary quality (Columns 3–4). Second, we added the interaction term between the paved village road and the 5V7H connection in the original Eq. (1) and re-estimate it, estimated coefficients and significance levels of the 5V7H connection remained substantially the same as the base results. Moreover, compared with rural residents from villages without paved roads, those from villages with paved roads benefit more from the 5V7H connection in terms of DDS and CHEI (Columns 7–8). These findings imply a complementary relationship between the 5V7H connection and paved village roads in improving dietary quality among rural residents.

Placebo tests: We followed Cai et al. (2016) and Ferrara et al. (2012) and randomly assigned 5V7H connection and connection years to conduct placebo tests, respectively. As mentioned above, of the 120 sample villages, 34 villages are connected to the 5V7H in one of the four years: 1997, 2000, 2004, and 2006. Therefore, we first randomly assigned the "placebo" 5V7H connection to 34 villages and constructed a dummy variable $\text{connection}_{vt}^{false}$ that takes the value of one for "placebo" treatment villages and zero otherwise. Then, for each of the 34 "placebo" treatment villages, we randomly assigned one of the four connection years (1997, 2000, 2004, and 2006) to be its "placebo" connection year. With this, we constructed a "placebo" post-connection year dummy variable post_{vt}^{false} that takes the value of one for years post the "placebo" connection year and zero otherwise. Finally, we took the interaction term of $\text{connection}_{vt}^{false}$

$*\text{post}_{vt}^{false}$ and specified Eq. (4) as follows.

$$y_{jit} = \alpha^{false} + \beta^{false}\text{connection}_{jvt}^{false} * \text{post}_{jvt}^{false} + \gamma^{false} Z_{jit} + \mu_j + \theta_m + \rho_t + \gamma_{jt} + \varepsilon_{jit} \tag{4}$$

As both the "placebo" 5V7H connection and the "placebo" connection year are randomly assigned, this newly constructed interaction item should have no effect on dietary quality (i.e., $\beta^{false} = 0$). The placebo estimates from the 500 randomized assignments show that the distribution is centered around zero with a p-value bigger than 0.10 for most outcomes (Appendix Fig. 2), suggesting it is unlikely that the positive and significant effects of the 5V7H connection on rural residents' dietary quality are driven by unobserved factors. Overall, results from these placebo tests lend further credence to our staggered DID identification strategy.

6.4. *Mechanism analyses*

6.4.1. *Exploring potential mechanisms*

The results reported above show that rural residents connected to the 5V7H have better dietary quality in terms of both DDS and CHEI. Why is it like this? We follow the conceptual framework developed above to examine the potential mechanisms underlying these findings and present the results in Tables 7 and 8.

Improved access to markets. We measure access to markets by the average distance from village committee to nearby food stores.[13] This average distance measure was derived from business registration data issued by the State Administration for Industry and Commerce (SAIC), which includes detailed information on registration, geographic location, and industry classification for all formal enterprises in China (Dai et al., 2021; Shi et al., 2021). We then replace the left hand side variable in Eq. (1) with the average distance measure to examine the impact of the 5V7H connection on access to food market.[14] The results show that compared to villages without 5V7H connection, villages with such connection were 2.625 km closer to the nearest food store, 3.359 km closer to the average of the nearest three stores, and 4.377 km closer to the average of the nearest five stores (Columns 1–3 in Table 7). These findings suggest that the 5V7H connection improved rural households' access markets.

Moreover, we explore whether the 5V7H connection affected local food prices. To do so, we replace the left hand side variable in Eq. (1) with local food prices and reran it.[15] The results show that the 5V7H connection significantly reduced prices for rice, flour, and vegetables (Columns 4–17, Table 7). Prices of fish, meat, and cooking oils also declined, although the point estimates are not statistically significant. One exception is eggs, for which prices increased following the 5V7H connection. This may reflect improved market integration: eggs produced locally could have been redirected to more profitable urban markets, thereby reducing local supply and pushing up prices in rural areas.

Better off-farm employment and income opportunities. We replace the left-hand side variable in Eq. (1) with the share of household members engaged in off-farm employment and per capita household income to test the impact of the 5V7H connection on off-farm employment and income opportunities, respectively.[16] Regression results show that, compared to their peers without 5V7H connection, rural households with connection to the 5V7H have 4.75 percentage points more household members engaged in off-farm employment (Column 1 in Table 8) their per capita household income is 940 yuan higher (Column 2).

Improving dietary literacy. We follow Conley and Udry (2010) and Maertens et al. (2020) to employ the "information acquisition-knowledge updating-behavioral change" approach to test whether the 5V7H connection helps to improve the dietary literacy of rural residents. Specifically, we replace the left-hand side variable in Eq. (1) with the following three indicators, one at a time, a.) the pieces of information acquisition tools (e.g., TV, mobile phone, and computer); b.) dietary knowledge; and c.) cooking patterns (frying, stir-frying, steaming, boiling, etc.). For details on constructions of these variables, please refer to Appendix A2. Regression results show that the 5V7H connection significantly increased the pieces of information acquisition tools of rural residents and their dietary knowledge (Columns 3–5). Moreover, we also found that the 5V7H connection contributes to a shift in meal preparation from unhealthy to healthy patterns (Column 6).

Other Potential Mechanism. We also examined two additional mechanisms: refrigerator ownership and the transition toward high-value agriculture. Column 1 of Appendix Table 2 presents the estimated effect of the 5V7H connection on the number of refrigerators per household. Although the coefficient is positive, it is not statistically significant. Moreover, we find no evidence that the 5V7H connection led to a shift toward high-value agricultural production (Columns 2–5 of Appendix Table 2).

6.4.2. *Mediation analysis*

So far we have examined both supply- and demand-side mechanisms through which the 5V7H connection may have improved dietary quality in terms of DDS and CHEI. To directly link each mechanism to the observed dietary changes, we conduct mediation analysis to quantify the relative contributions of each channel. As discussed above, the demand-side mediators include the share of

[13] Specifically, we first sorted all registered businesses by year and operational status, keeping only those that were in operation during each CHNS survey year (1997, 2000, 2004, 2006, and 2009). From this subset, we then referred to the National Industry Codebook to identify food-related retail establishments by focusing on those registered business that are coded F521 ("General retail trade"), or F522 ("Specialized retail trade of food, beverages and tobacco products"). Finally, based on the geographic coordinates of these food-related retail businesses, we calculated the straight-line average distances from each village seat to the nearest one, three, and five food retail stores.

[14] For more details on estimation strategy, please refer to Appendix A1.

[15] For more details on estimation strategy, please refer to Appendix A1.

[16] Income is adjusted using the GDP deflator with 1978 as the base year.

Table 9
Mediation analysis: supply-side mechanism.

	(1)	(2)	(3)	(4)	(5)	(6)
	Mediating Variables: Distance to Food Stores					
	Nearest		3-Nearest Avg.		5-Nearest Avg.	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
Panel A. Dietary Diversity Score (DDS)						
Indirect Effect	0.059**	0.030	0.053*	0.030	0.054*	0.059
Direct Effect	0.271**	0.112	0.278**	0.111	0.276**	0.271
Total Effect	0.330***	0.111	0.330***	0.111	0.330***	0.330
Proportion of indirect effect	0.179		0.159		0.163	
Panel B. Chinese Healthy Eating Index (CHEI)						
Indirect Effect	0.032	0.073	0.000	0.060	−0.012	0.064
Direct Effect	2.218***	0.793	2.250***	0.787	2.262***	0.783
Total Effect	2.250***	0.788	2.250***	0.788	2.250***	0.788
Proportion of indirect effect	0.014		0.000		−0.005	

Notes: Panel A and Panel B report results using the Dietary Diversity Score (DDS) and the Chinese Healthy Eating Index (CHEI) as dietary quality indicators, respectively.
All regressions control for the individual/household/regional characteristics as discussed above, year fixed effects, county fixed effects, and survey month fixed effects; standard errors are clustered at the county-year level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

household members employed off-farm, per capita household income, the number of information tools, and cooking patterns, while the supply-side mediator includes the distance from the village committee to the nearest food store. Due to data limitations, we have to ignore dietary knowledge as it is only available after 2004. Food prices were also ignored as out of the 13 food groups, many had missing values, which makes it impossible to combined them into an index. Results from the mediation analysis show that the 5V7H connection enhances rural residents' dietary quality primarily through demand-side pathways rather than supply-side improvements.

Supply-side factors. The distance to the nearest food store significantly mediates the effect on DDS, accounting for 17.9 percent of the total effect, but no significant mediating effect is found for CHEI (Table 9). This indicates while closer proximity to food stores may improve dietary diversity, it does not necessarily lead to a healthier overall dietary composition.

Demand-side factors. Specifically, the share of household members employed off-farm significantly mediates the effect, accounting for 9.3 percent of the total effect on DDS whereas 4.4 percent on CHEI; the number of information tools also has a significant mediating effect, explaining 9.6 percent of the total effect on DDS and 5.6 percent on CHEI (Columns 1–4 in Table 10).

Per capita household income and cooking pattern act as mediators with variations due to their nonlinear relationships with dietary quality (Fig. 6).[17] For individuals with per capita household income above 1600 CNY (25th percentile), the mediating effects account for 4.6 percent of DDS and 2.20 percent of CHEI. For those with incomes over 3000 CNY (50th percentile), the effects decline to 3.0 percent for DDS and become insignificant for CHEI, with no significant mediation observed beyond 5500 CNY (75th percentile). For those with frying and stir-frying ratios above 23.95 percent (25th percentile), the mediating effects are 5.5 percent for DDS but insignificant for CHEI. When the frying and stir-frying ratio exceeds 35.97 percent (50th percentile), the mediating effects increase to 10.30 percent and 5.50 percent. For those with the ratio over 46.80 percent (75th percentile), the mediating effects increase to 14.10 percent and 6.60 percent, respectively (Columns 5–16, Table 10). Appendix Table 3 also provides supporting evidence showing that high-income households are more likely to exhibit high frying and stir-frying ratio than their lower-income counterparts. This evidence suggests that the 5V7H connection improves dietary quality through distinct pathways across income groups: low-income groups benefit more from the income channel while high-income groups gain primarily through the dietary literacy channel.

7. Heterogeneous effects of the 5V7H connection on rural residents' dietary quality

There is evidence that the effects of better transportation on dietary quality vary by household characteristics (Hirvonen et al., 2017; Stifel and Minten, 2017). In this sub-section, we examine the potential heterogeneous effects of the 5V7H connection on rural residents' dietary quality by repeating analyses reported in Table 11 with additional interaction terms in Eq. (1). We focus on five household characteristics: the diversity of agricultural production measured by the number of agricultural productions that households are engaged in (Huang and Tian, 2019) ;[18] refrigerator ownership measured by a dummy variable that takes the value of one for households possessing refrigerators and zero otherwise; the number of household children aged 2–14 years; and the number of elderly household members aged 60 years and above; the education of household meal preparer (in years).

Results from heterogeneous effects of the 5V7H connection on dietary quality reveal some informative patterns (Table 11). First, we

[17] The sample is divided into subgroups based on per capita household income levels above 1,600, 3,000, and 5,500 CNY, corresponding to the 25th, 50th, and 75th percentiles of the income distribution. Similarly, subgroups are defined according to frying–stir-frying ratios above 23.95%, 35.97%, and 46.80%, corresponding to the 25th, 50th, and 75th percentiles of the distribution.
[18] The CHNS collects information on four sources of household income related to agriculture as follows, cropping, fishing, horticulture, and livestock.

Table 10
Mediation analysis: demand-side mechanisms.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	Mediating variables:															
	Share of household labor off-farm (%)		Number of information tools		Per capita household income (×10,000 CNY)						Frying and stir-frying ratio (%)					
	/		/		Over 0.16 (25%)		Over 0.30 (50%)		Over 0.55 (75%)		Over 23.95 (75%)		Over 35.97 (50%)		Over 46.80 (25%)	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
Panel A. Dietary Diversity Score (DDS)																
Indirect Effect	0.031*	0.017	0.031*	0.018	0.014**	0.007	0.011*	0.006	0.006	0.005	0.018**	0.008	0.034**	0.013	0.050**	0.024
Direct Effect	0.300***	0.110	0.293***	0.109	0.299**	0.120	0.366***	0.127	0.126	0.151	0.303***	0.111	0.300**	0.130	0.305*	0.174
Total Effect	0.330***	0.111	0.324***	0.108	0.313**	0.120	0.377***	0.126	0.132	0.150	0.320***	0.112	0.334**	0.132	0.354**	0.180
Proportion of indirect effect	0.093		0.096		0.046		0.03		0.043		0.055		0.103		0.141	
Panel B. Chinese Healthy Eating Index (CHEI)																
Indirect Effect	0.099*	0.057	0.129*	0.073	0.043*	0.026	0.030	0.026	0.004	0.032	0.040	0.040	0.135**	0.062	0.187*	0.096
Direct Effect	2.151***	0.775	2.160***	0.788	1.924**	0.920	2.425**	0.981	1.743*	0.933	2.172***	0.825	2.313***	0.860	2.632**	1.075
Total Effect	2.250***	0.788	2.289***	0.790	1.967**	0.919	2.455**	0.982	1.747*	0.934	2.213***	0.83	2.448***	0.866	2.819**	1.090
Proportion of indirect effect	0.044		0.056		0.022		0.012		0.003		0.018		0.055		0.066	

Notes: Panel A and Panel B report results using the Dietary Diversity Score (DDS) and the Chinese Healthy Eating Index (CHEI) as dietary quality indicators, respectively.
Due to the nonlinear relationships between per capita household income, frying and stir-frying ratio, and dietary quality, mediation effects were estimated using different subgroups of the sample (Columns 5–16). Thresholds for per capita household income were set at 1500, 3000, and 5500 CNY, and for the frying and stir-frying ratio at 23.95%, 35.97%, and 46.80%, corresponding to the 25th, 50th, and 75th percentiles of the respective distributions.
All regressions control for the individual/household/regional characteristics as discussed above, year fixed effects, county fixed effects, and survey month fixed effects; standard errors are clustered at the county-year level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

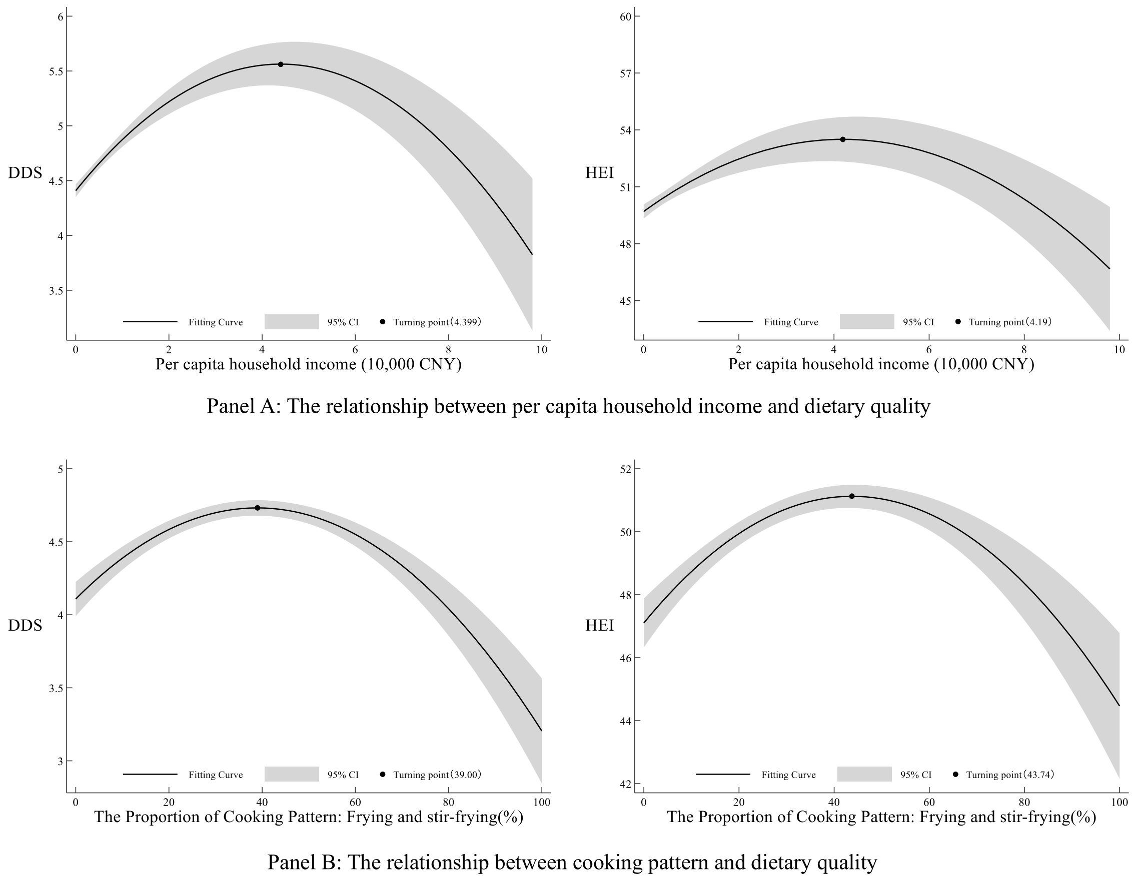

Fig. 6. The relationship between household income, cooking patterns, and dietary quality.
Notes: The figure presents the fitted curves, 95% confidence intervals and turning point illustrating the relationships between per capita household income, cooking patterns, and dietary quality.

find that compared with rural residents from low agricultural production diversity households, those from high agricultural production diversity households benefit less from the 5V7H connection in CHEI but not in DDS. One possible explanation is that smallholder farmers typically consume a sizeable part of what they produce, which tends to be limited in nutrient composition (Sibhatu and Qaim, 2018b). Second, rural residents from households with refrigerator benefit more from the 5V7H connection in both DDS and CHEI. Third, rural residents from households with more children aged 2–14 years benefit more from the 5V7H connection in both DDS and CHEI. Finally, rural households with better educated meal preparers benefit more in DDS and CHEI from the 5V7H connection. In contrast, we find no evidence of heterogeneity effects by the number of elderly members (aged 60+).

8. Conclusions

In this paper, we have examined whether having better road transportation would benefit dietary quality for rural residents in developing countries. By exploiting the staggered roll-out of the “Five Vertical and Seven Horizontal” National Trunk Highway System (5V7H) in China, we took a staggered DID identification strategy and found statistically significant positive impacts of better transportation on rural residents’ dietary quality. The estimated effects imply an increase of about 0.33 in the number of food groups consumed over the three-day reference period and a meaningful improvement in alignment of rural residents’ diets with the Dietary Guidelines for Chinese. This improvement is further reflected in more consumption of recommended food groups, such as fruits, aquatic products, nuts, and dairy products, together with reduced oil intake.

We further investigated the mechanisms underlying these findings and provided empirical evidence that the 5V7H connection contributes to better off-farm employment opportunities, higher household income, improved dietary literacy, enhanced market access, and lower food prices. Results from the mediation analyses further reveal that the improvement in dietary quality is driven mainly by demand-side channels (better off-farm employment opportunities, higher household income, improved dietary literacy), while the contribution of supply-side factors is relatively limited (market access and food prices). Moreover, we also find evidence suggesting that the 5V7H connection improves dietary quality through distinct pathways across income groups: low-income households benefit more through the income channel, whereas high-income households gain more through the dietary literacy channel. Taken together, these findings indicate that the effects of the 5V7H connection on rural residents’ dietary quality stem primarily from enhanced dietary

Table 11
Heterogeneity effects.

	(1)	(2)	(3)	(4)	(5)
Panel A. Dietary Diversity Score (DDS)					
5V7H Connection	0.387***	0.231**	0.276**	0.324***	0.202*
	(0.120)	(0.112)	(0.108)	(0.112)	(0.114)
5V7H Connection × Agricultural production diversity (range 0–4)	−0.049				
	(0.035)				
5V7H Connection × Refrigerator ownership (1=Yes; 0=No)		0.188**			
		(0.086)			
5V7H Connection × Number of children			0.109**		
			(0.048)		
5V7H Connection × Number of elderly members				0.001	
				(0.047)	
5V7H Connection × Education of meal preparer (year)					0.022**
					(0.009)
Observations	18,893	18,893	18,893	18,893	18,525
R^2	0.329	0.345	0.320	0.318	0.321
Panel B. Chinese Healthy Eating Index (CHEI)					
5V7H Connection	3.226***	1.634**	2.003**	2.431***	1.403*
	(0.851)	(0.810)	(0.794)	(0.799)	(0.837)
5V7H Connection × Agricultural production diversity (range 0–4)	−0.725***				
	(0.218)				
5V7H Connection × Refrigerator ownership (1=Yes; 0=No)		1.276**			
		(0.561)			
5V7H Connection × Number of children			0.500*		
			(0.302)		
5V7H Connection × Number of elderly members				−0.333	
				(0.247)	
5V7H Connection × Education of meal preparer (year)					0.150***
					(0.054)
Observations	18,893	18,893	18,893	18,893	18,525
R^2	0.171	0.176	0.176	0.170	0.170

Notes: Panel A and Panel B report results using the Dietary Diversity Score (DDS) and the Chinese Healthy Eating Index (CHEI) as dietary quality indicators, respectively.
The interaction terms in Panel B are defined the same as in Panel A.
All regressions control for the individual/household/regional characteristics as discussed above, provincial trends, year fixed effects, county fixed effects, and survey month fixed effects.Standard errors clustered at the county-year level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
The regressions are weighted by the province-level sample size.

demand rather than improvements in dietary supply.

Results from the heterogeneity analyses show that the 5V7H's beneficial impacts on dietary quality are more prominent among rural households exhibiting three key characteristics: more children, refrigerator ownership, and meal preparers with better education, while demonstrating comparatively attenuated effects for households with more diversified agricultural production.

We acknowledge at least two limitations of our study due to data constraints. First, our identification relies on a simplified canonical DID model with just two periods (pre- and post-2004) when implementing the instrumental variable (IV) strategy based on Faber (2014). This approach sacrifices the rich temporal variation embedded in the staggered rollout of the 5V7H network. We use the IV strategy as a robustness check rather than a substitute for the DID approach. Nevertheless, the IV results provide complementary evidence by alleviating endogeneity concerns related to highway placement, even if within a more constrained empirical framework. Second, we were unable to conduct a comprehensive mediation analysis that includes key potential mediators such as changes in local food prices on the supply side and dietary knowledge on the demand side. Nevertheless, our findings still offer supporting evidence: the main channel through which the 5V7H connection improves dietary quality appears to lie in increased demand. While greater food availability may enhance dietary diversity, it does not significantly improve dietary health.

Despite these limitations, we believe our research findings shed light on policies that seek to improve the dietary quality of rural residents. First, when evaluating the benefits of road transportation, indirect benefits such as dietary health benefits should also be considered. For developing countries committed to improving the dietary quality of rural residents, road infrastructure may be one of the options. Second, we find that the 5V7H connection improves rural residents' dietary quality primarily through demand-side channels. This implies that policies aiming to promote healthier diets in developing countries should focus on stimulating demands such as through improved education on dietary and nutritional knowledge. At the same time, our findings underscore the role of road infrastructure in enabling employment opportunities. This highlights the need for complementary policies alongside infrastructure investment, including greater access to labor market information and training in off-farm employment skills. Finally, we found that the 5V7H has less impact on households with more diversified agricultural production, which implies dietary policies should be formulated with more attention to these groups.

CRediT authorship contribution statement

Xuyuan Zheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Qinglin Lin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Gang Xie:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Xiaohua Yu:** Writing – review & editing, Resources, Data curation, Conceptualization. **Chengfang Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of interest statement

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jhealeco.2026.103139.

Data availability

The authors do not have permission to share the data.

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